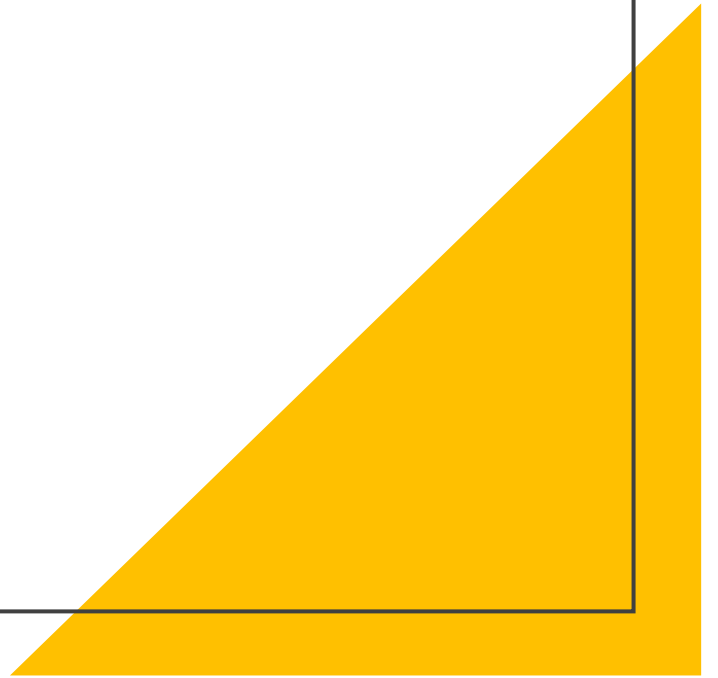


Getting started in Research and Analytics

Session 1: Distributions & Correlations



Syllabus: Getting started in Research and Analytics (with SPSS, R, & Python)

Purpose: An introduction to each topic will help answer two questions, “What is it?” and “How do we run it with statistical software?”.

The focus of the sessions will be on understanding key concepts and using software to calculate statistical tests and produce graphs to help better understand our study questions.

The software options we will review are SPSS, R, and Python. Programming code will be made available whenever possible. I’ll use R Studio and VS Code (Python).

Presentations*

1. Distributions and Correlations
2. Statistical Inference & Hypothesis Testing
3. Power Analysis
4. Survey Research Methods
5. Linear & Logistic Regression
6. Matching

* Please see Appendix A for the course outline

Notes: All presentation notes will be provided in power point files. Stats books are in the StatsWithSteve channel.

Code: SPSS, R, and Python code will be made available whenever possible.

Data: Data are in CSV format and can be opened with SPSS, R, and Python.

Files: All power points, code, and data are found in the StatsWithSteve channel or Github.

<https://github.com/szuniga07/Statistics/tree/main/Getting%20Started%20in%20Research%20%26%20Analytics>



Distributions and Correlations

In this session,
we'll cover
Distributions and
Correlations.

Statistical tests
and graphs help
us explain our
data.

Why should we learn this?

Distributions give us the center and spread. ***Variation = uncertainty.***

Correlations tell us about important relationships in the data. ***X -> Y***



Research and Analytics

- In research, we use statistics to describe our data and make inferences to the larger population with the goal of “generalizability”.
 - In analytics, we use statistics to do exploratory analyses to help guide teams on where to focus, to make judgments under uncertainty. Nobel prize winner Kahneman (2002), stresses the importance of using data in decision making.
 - The main difference is that we test hypotheses in research. However, there can be a lot of overlap in research and analytics, we use identical statistics.
-

Software

Each software has advantages:

- SPSS is very user friendly.
- R is a powerful statistical tool.
- Python works well for data management.
- All 3 do the same work but different users may want to begin with a specific software.
- Open-source R and Python are flexible and **free**. SPSS has a long history of its product (e.g., textbooks), runs code, and has customer support.
- There are many good sources on installing the software, etc. We'll focus on demonstrating how to use the software for various analyses.

Data

- We'll use a hypothetical dataset, “hosprog” from the R package, ham.
- Data is available in a CSV file to upload into each software.

Distributions

Statistical tests and SPSS graphs

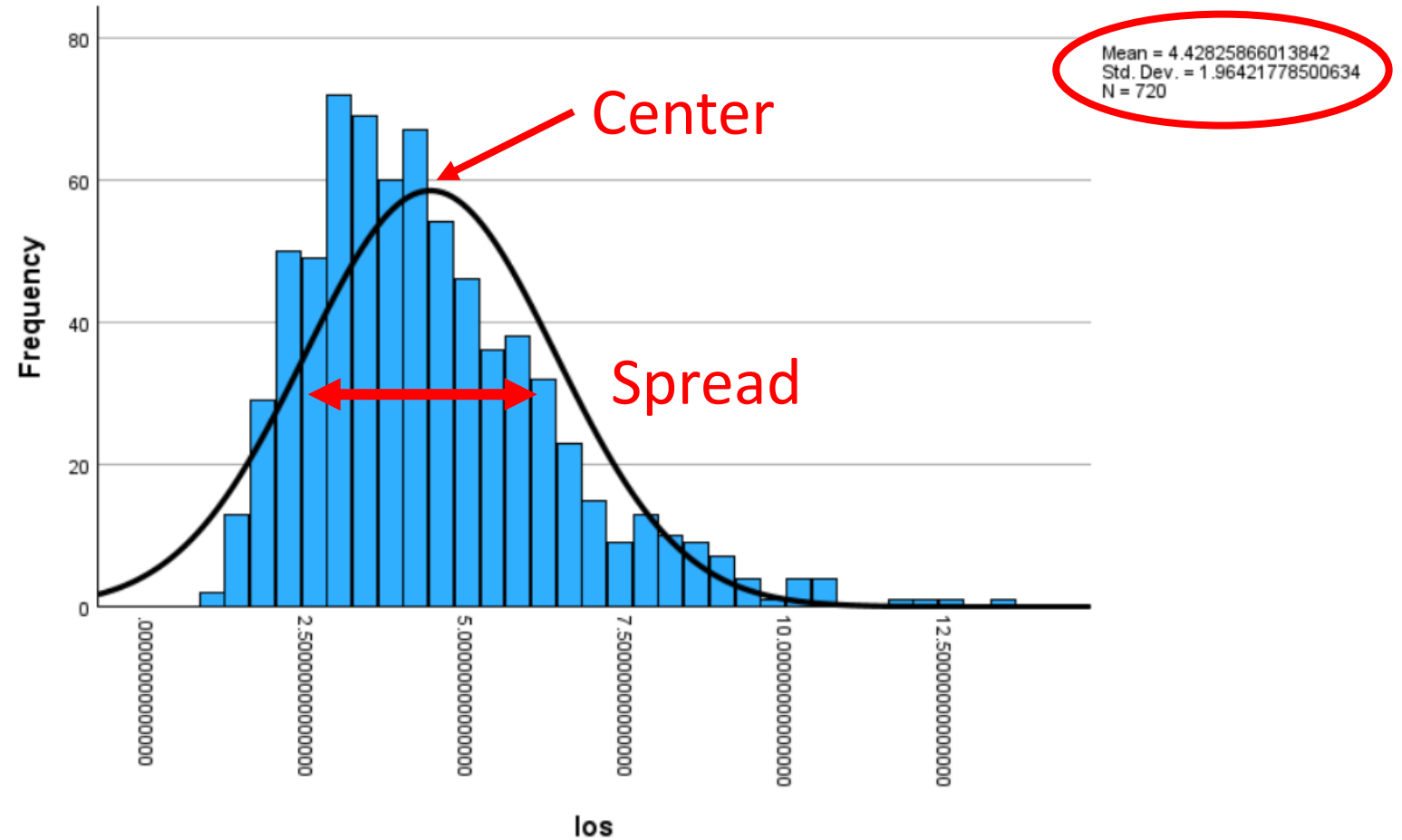
Histogram of Length of Stay with a Normal Curve

Histograms have bars that indicate the frequency of responses within a range of values.

For example, about 50 patients had LOS of 2, 3, 4, or 5 days.

The black bell-shaped line is the normal curve based on the mean and standard deviation in the top right. Doesn't line up with data... "that's ok".

This graph was created in SPSS, see Appendix B to learn how (with all other SPSS examples).

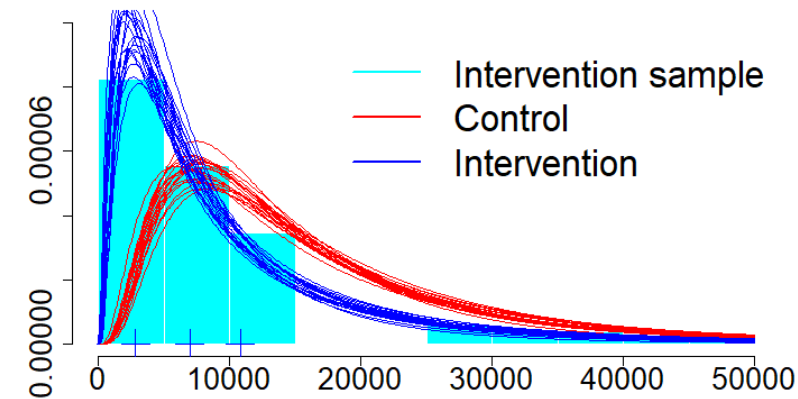


We always report the average ("center") but why should we discuss the "spread"?

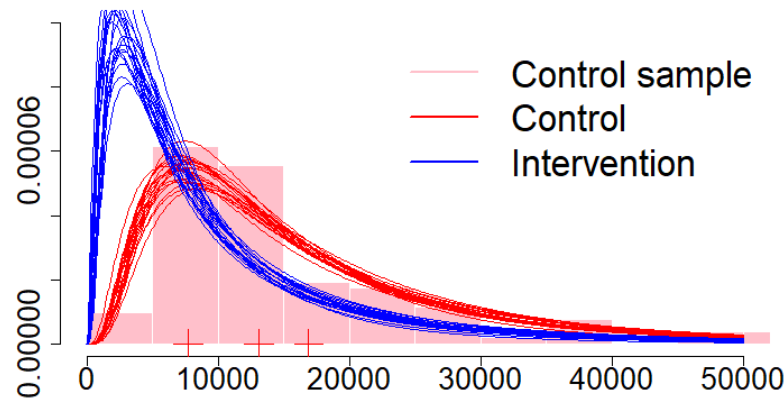
- Variation or spread tells us how certain we are about the average. Wouldn't it be nice to say the average LOS is 4 to 5 days instead of 0 to 10 days?
- More variation/spread = More "uncertainty".

Distributions reveal important results (Bayesian Estimation)

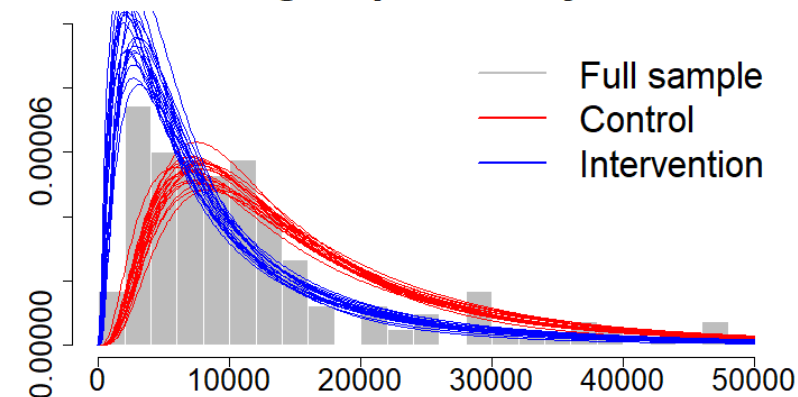
Treatment's 30-day cost



Control's 30-day cost



Both group's 30-day cost

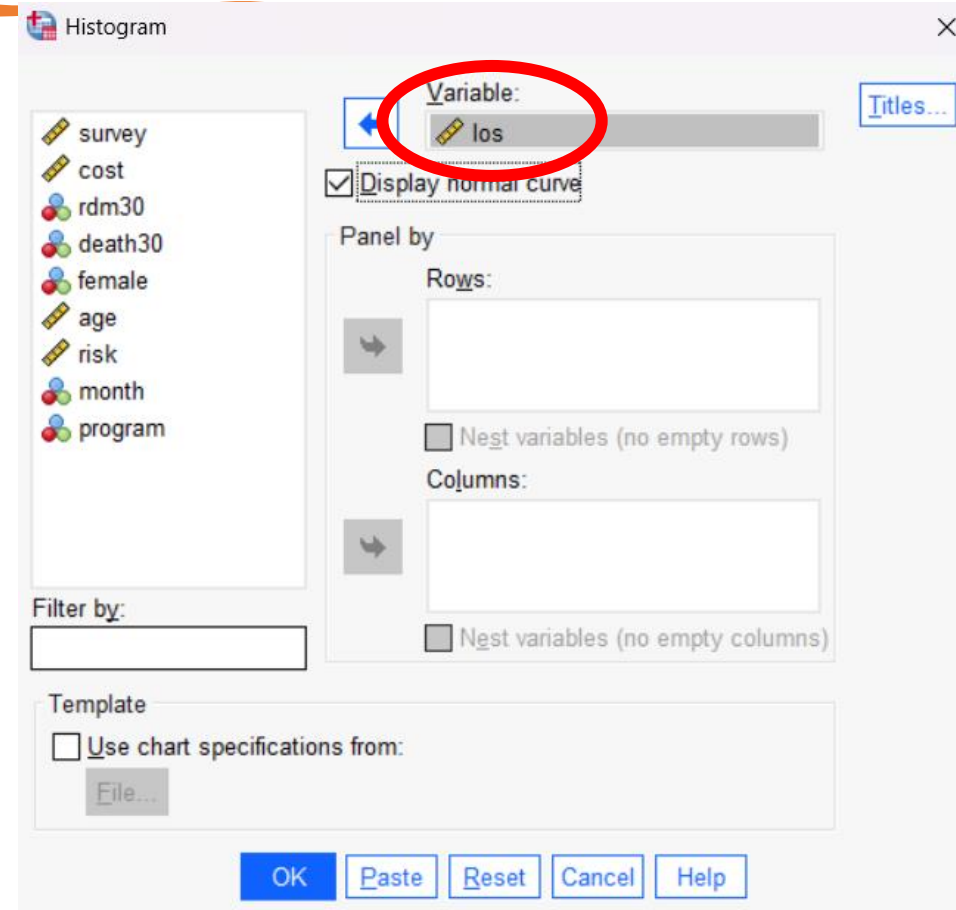
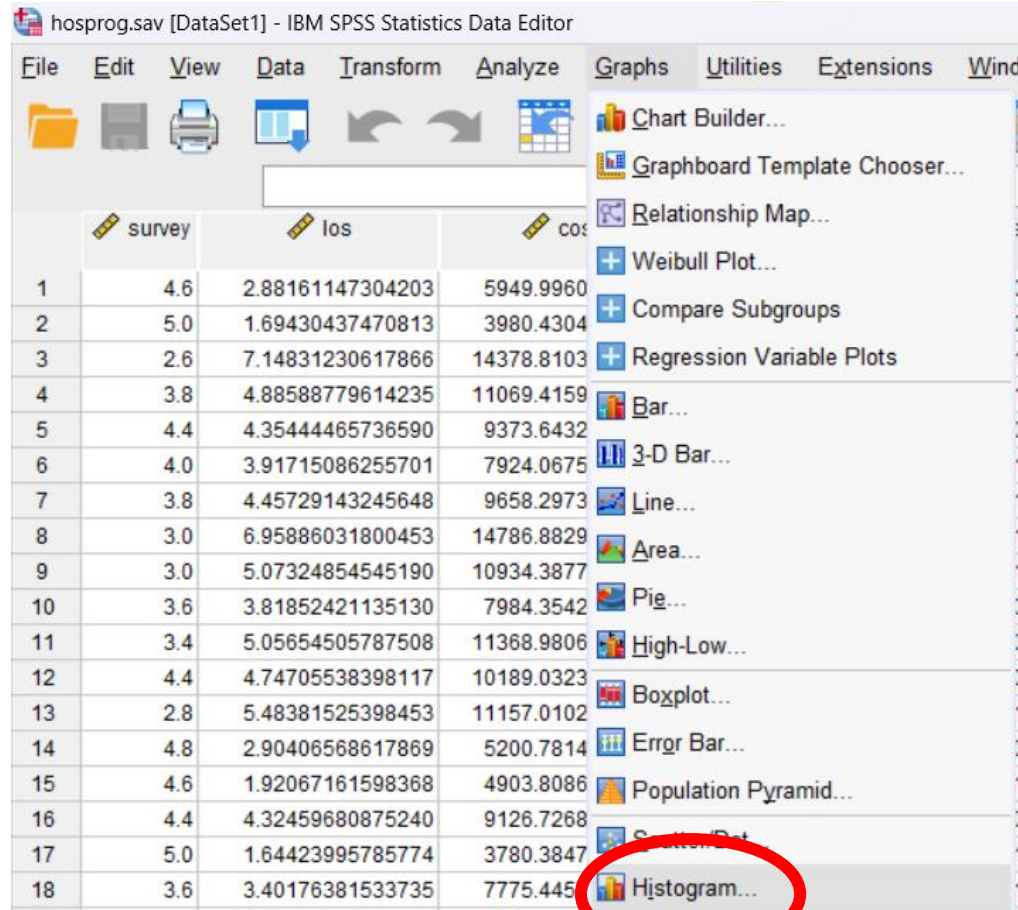


Mayo Clinic teamed up to introduce care at home nationwide. See Mayo doctors describe equally safe care and reduced costs on "The Today Show".

<https://www.today.com/video/new-technology-provides-hospital-level-care-from-patients-homes-119185989747>

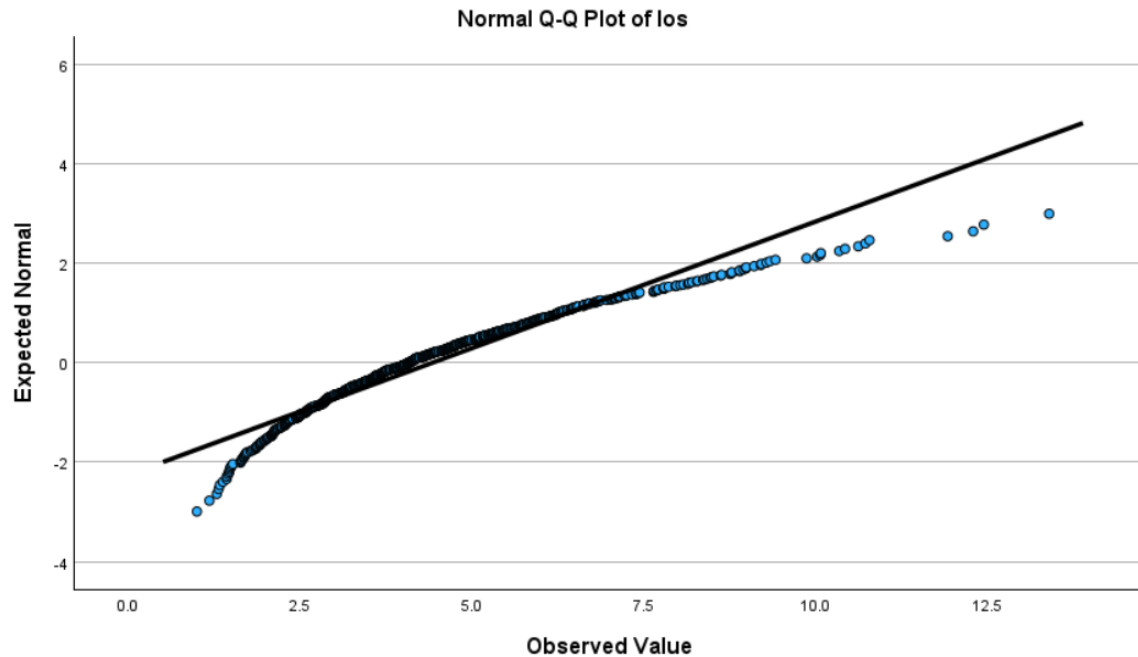
- **Care at Home**
- Groundbreaking program, based on Johns-Hopkins. Provides equally safe and cost-effective care at home. Covid care.
- Top-left: Light-blue bars and dark blue lines show hospital costs lower than traditional care.
- Top-right: Pink bars and red lines show higher traditional care costs.
- Bottom: The red and blue distributions clearly show 2 different experiences in terms of cost in the full matched data.
- Try it with Density Plots.

Creating a Histogram



Alternative checks: Q-Q plot and Kolmogorov-Smirnov test

These are ways to check if our sample's data is normally distributed...but is a normal distribution required? **No.**



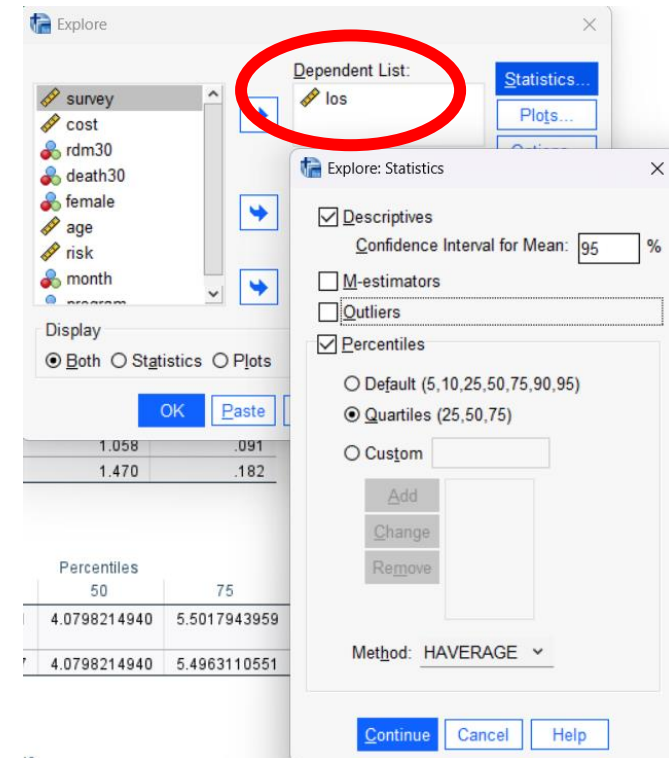
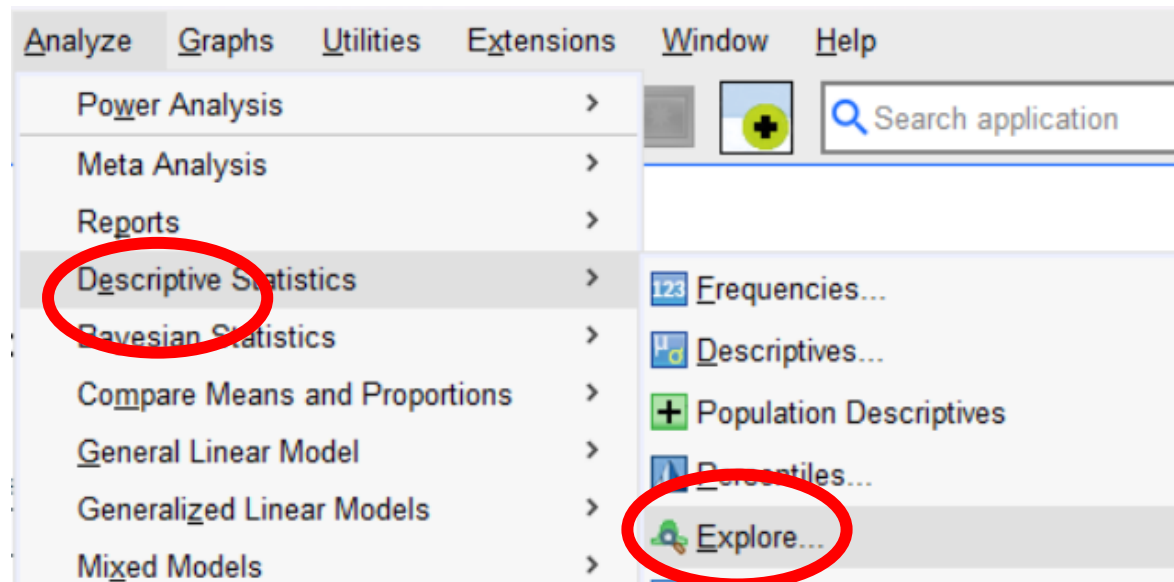
- The flatness on the right side (more horizontal) suggests more of a tail on the right or positively skewed data.
- The steeper increase on the left side (more vertical) suggests more data in that part of the distribution.
- In the middle, the data falls along the theoretical quantiles so it looks more normal.

Tests of Normality						
	Statistic	Kolmogorov-Smirnov ^a		Statistic	Shapiro-Wilk	
		df	Sig.		df	Sig.
los	.086	720	<.001	.938	720	<.001

a. Lilliefors Significance Correction

When Sig is < 0.05 , the result is significant and indicates the shape is significantly different from normal. *This test is probably easier to use than the Q-Q plot.*

How to get Descriptive Statistics



Descriptive Statistics on the Center and Spread of the Distribution

Descriptives

			Statistic	Std. Error
los	Mean		4.4282586601	.07320207483
	95% Confidence Interval for Mean	Lower Bound	4.2845433065	
		Upper Bound	4.5719740138	
	5% Trimmed Mean		4.2976484466	
	Median		4.0798214940	
	Variance		3.858	
	Std. Deviation		1.9642177850	
	Minimum		1.0000000000	
	Maximum		13.392649075	
	Range		12.392649075	
	Interquartile Range		2.5276030338	
	Skewness		1.058	.091
	Kurtosis		1.470	.182

Percentiles give us the % of the sample that is below a certain point.

For example, the 25th percentile is 2.97, 25% of the sample has values lower than that.

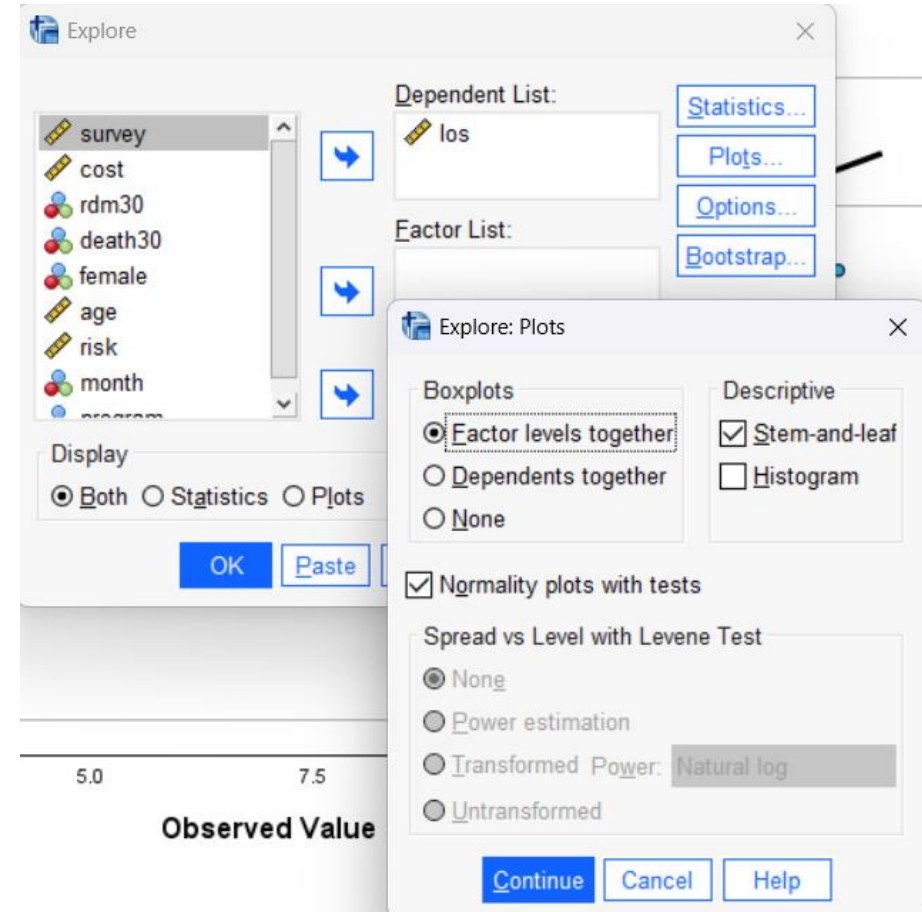
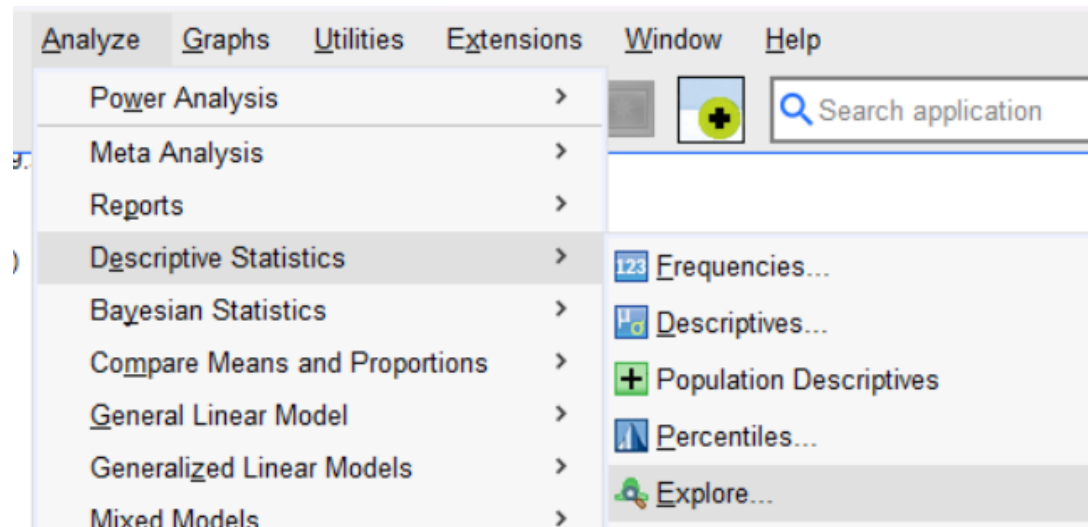
Percentiles

		25	Percentiles 50	75
Weighted Average (Definition 1)	los	2.9741913621	4.0798214940	5.5017943959
Tukey's Hinges	los	2.9768850727	4.0798214940	5.4963110551

Quartiles are the 25th, 50th, and 75th percentiles

The Standard Deviation is the square root of the Variance. In a normal distribution, the 68-95-99.7 rule is that 68%, 95%, and 99.7% of the sample falls within 1, 2, and 3 standard deviations around the mean.

How to create a Q-Q plot and run a Kolmogorov-Smirnov test



Summary



- A visual inspection of a histogram helps understand normality.
- Key parts of the distribution are the “center” and “spread”, often referred to as the mean and standard deviation.
- Understanding the distribution allows us to make sense of what is going on with the outcome and ideas of how other factors impact it.

Correlations and Scatterplots

Statistical tests and SPSS, R, and Python results

Different reasons for using correlations

Reasons



Check to see if 2 variables are “related”.



Check to see if there is a trend in data.



Reliability of one variable as a measure of another variable.



Make predictions.



Create composite variables when R is high (≥ 0.95) or suitable.

Examples

1. Correlate yearly 30-day mortality and 30-day readmission rates.
2. Correlate months (e.g., 1-36) and LOS to see if we improved.
3. Correlate a gold standard med device with a cheaper option.
4. Correlate age and LOS .
5. The average of 2 survival scores or mean blood pressure.

Strength of correlations

A correlation is a rank-order comparison between 2 variables. For example, the highest ages are associated with the highest (or lowest) lengths of stay.

The correlation coefficient (R) ranges from -1 to 1. Negative and positive correlations. R=0 indicates no correlation.

The following rules apply to positive correlation values but have the same meaning for negative correlations. General guidelines in the strength of correlations: 1) ≥ 0.70 is strong, 2) $0.30 - 0.70$ is moderate, 3) < 0.30 is weak, 4) $0.0 =$ no correlation, 5) correlations of -1 or 1 are perfect correlations.

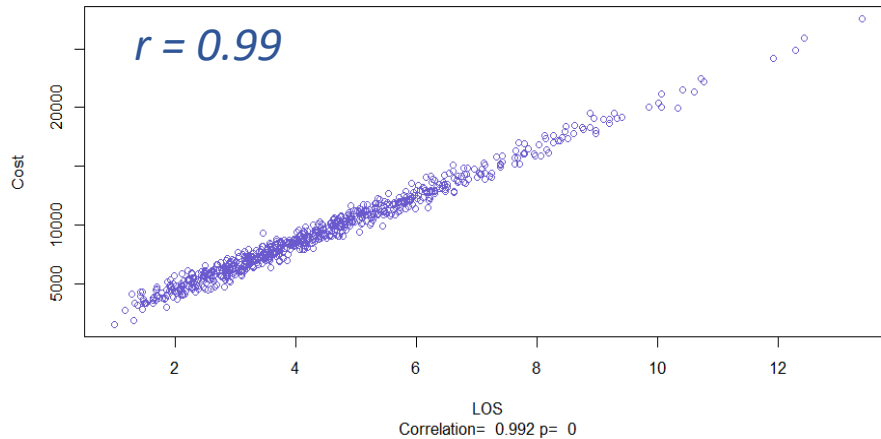
Causation vs. correlation vs. confounding: Causation implies a cause-and-effect path. Cause and effects can be correlated but correlations aren't always cause-and-effects. Confounders can be a 3rd variable correlated with both independent and dependent variables that impact relationships (e.g., motivation can confound the correlation between math ability and SAT math scores, **W** $\rightarrow$ X $\rightarrow$ Y).

Scatterplots let us see the relationships.

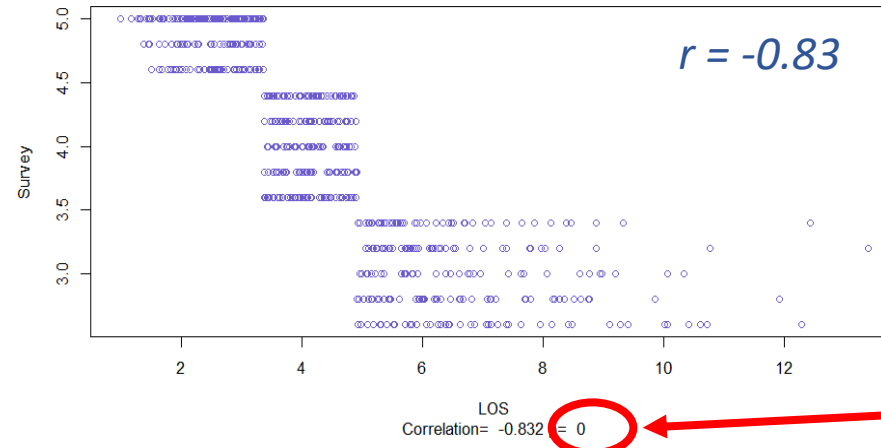
Correlation tests give us an idea on the magnitude.

All are very significant but tell us different things.

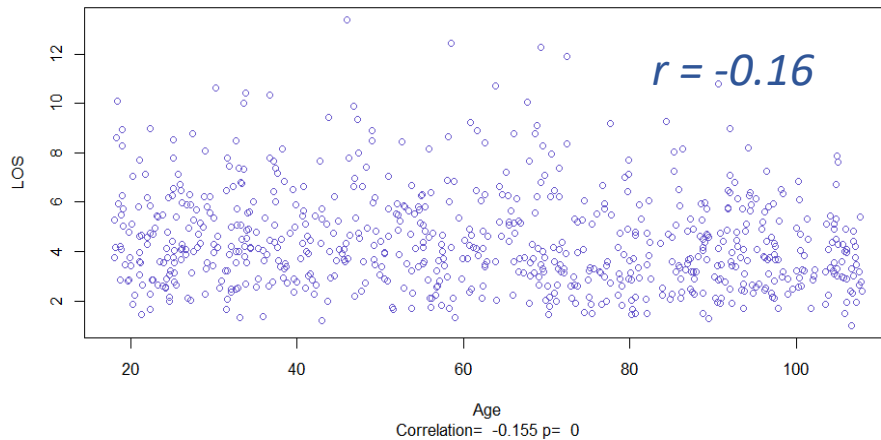
LOS x Cost (strong +)



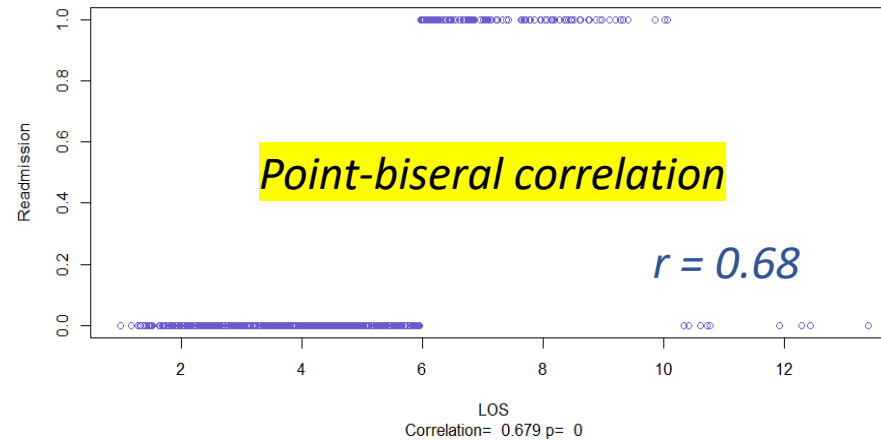
LOS x Survey (strong -)



LOS x Age (weak -)



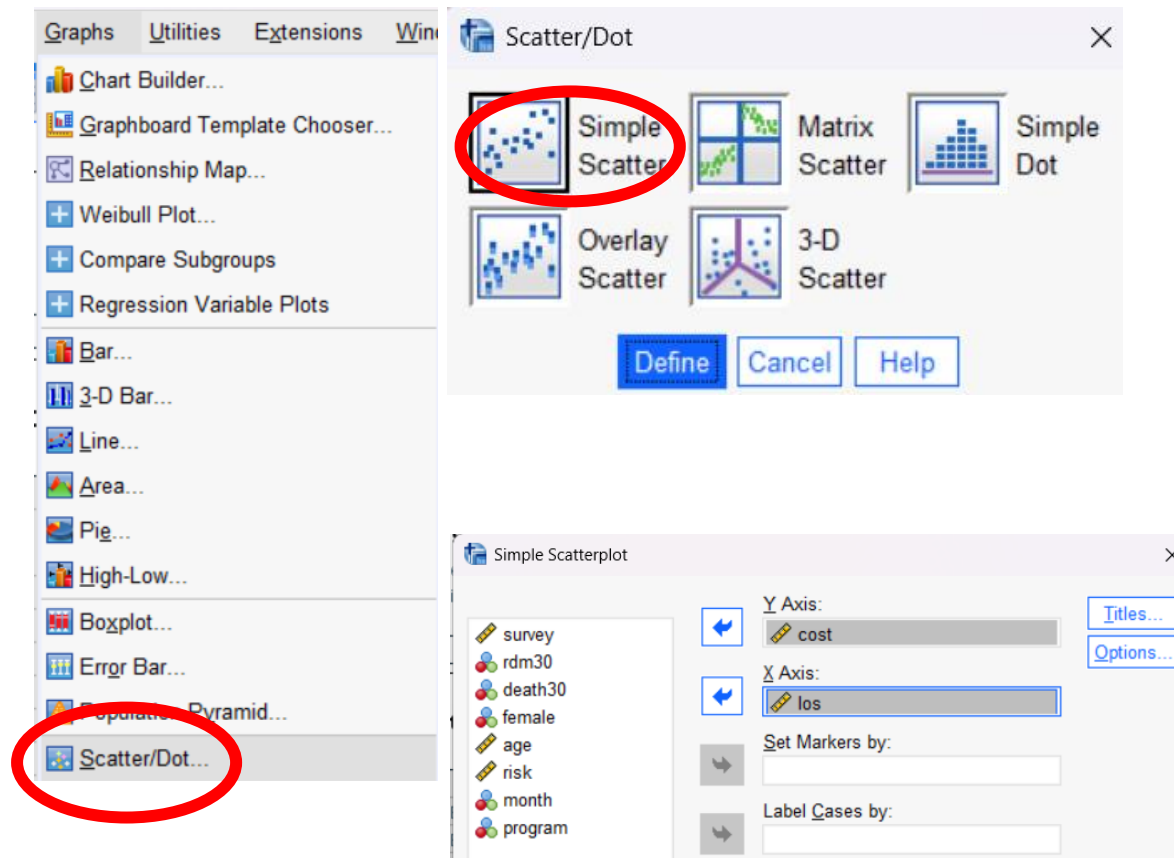
LOS x Readmission (moderate +)



Traditionally,
Outcome on y-axis,
Predictor on x-axis

SPSS and R code for LOS by Cost

SPSS



R

SPSS creates a very nice graph. R can create custom graphs with extra options in the standard code.

```
#1. Very strong positive correlation between LOS and cost
cor.test(hosprog$los, hosprog$cost, use = "complete.obs",
         method = "pearson" )

# Let's take a look
plot(hosprog$los, hosprog$cost )

# Adding "arguments" to give more info
plot(hosprog$los, hosprog$cost, main= "LOS x Cost (strong +)",
     xlab= "LOS", ylab= "Cost", col="cyan3",
     sub= paste("Correlation= ", round(cor1$estimate, 3), #paste text &
               "p= ", round(cor1$p.value, 3)) ) # numbers for subtitle
```

Correlation Matrix in SPSS and Python

SPSS

		Correlations					
		los	cost	survey	rdm30	female	age
los	Pearson Correlation	1	.992**	-.832**	.679**	-.052	-.155**
	Sig. (2-tailed)		<.001	<.001	<.001	.160	<.001
	N	720	720	720	720	720	720
cost	Pearson Correlation	.992**	1	-.827**	.674**	-.049	-.158**
	Sig. (2-tailed)	<.001		<.001	<.001	.193	<.001
	N	720	720	720	720	720	720
survey	Pearson Correlation	-.832**	-.827**	1	-.574**	.061	.170**
	Sig. (2-tailed)	<.001	<.001		<.001	.099	<.001
	N	720	720	720	720	720	720
rdm30	Pearson Correlation	.679**	.674**	-.574**	1	-.061	-.099**
	Sig. (2-tailed)	<.001	<.001	<.001		.102	.008
	N	720	720	720	720	720	720
female	Pearson Correlation	-.052	-.049	.061	-.061	1	-.024
	Sig. (2-tailed)	.160	.193	.099	.102		.528
	N	720	720	720	720	720	720
age	Pearson Correlation	-.155**	-.158**	.170**	-.099**	-.024	1
	Sig. (2-tailed)	<.001	<.001	<.001	.008	.528	
	N	720	720	720	720	720	720

** . Correlation is significant at the 0.01 level (2-tailed).

Python

There are identical values above and below the diagonal (where values are 1.000)

	survey	<u>los</u>	cost	rdm30	death30	female	age
survey	1.000	-0.832	-0.827	-0.574	-0.463	0.061	0.170
<u>los</u>	-0.832	1.000	0.992	0.679	0.343	-0.052	-0.155
cost	-0.827	0.992	1.000	0.674	0.334	-0.049	-0.158
rdm30	-0.574	0.679	0.674	1.000	0.448	-0.061	-0.099
death30	-0.463	0.343	0.334	0.448	1.000	-0.080	-0.053
female	0.061	-0.052	-0.049	-0.061	-0.080	1.000	-0.024
age	0.170	-0.155	-0.158	-0.099	-0.053	-0.024	1.000

Python code for the correlation matrix: Identical steps for 2 or more variables in SPSS

Import required libraries

```
#install once if needed  
#no need to re-run pip commands
```

```
pip install pandas  
import pandas as pd
```

Run the code

```
main_vars = ["survey",  
"los", "cost", "rdm30", "death30", "f  
emale", "age"] #select variables
```

```
hos_cor =  
hosprog[main_vars].corr()  
print(hos_cor.round(3))
```

*Correlation matrix created with Python, see file Week 1.py for all code.
"main_vars" and "hos_cor" are object names.*

Special considerations



Small sample statistic. Correlations are helpful with large and small samples. But we should be careful with aggregated data because the correlation coefficient can be higher than expected.



R^2 represents the amount of variance in Y (dependent variable) explained by X (independent variable). Knowing X gives us an idea of why we get different Y values. For example, when $R = -0.78$, the R^2 is 0.60. The Pearson correlation gives the same R^2 that we get from a simple linear regression.



Pearson, Spearman, and Kendall tau correlations can handle continuous, ordinal, and binary data. Kendall works especially well in ordinal correlations (e.g., survey questions), Spearman works well in skewed data. Pearson is the fastest to calculate results (e.g., big data). There are many more types of correlations.



Partial correlations control for other variables but we get more power in regression.

Summary



- Although it is true that “correlation doesn’t equal causation”, causation does equal correlation (i.e., composite variable if $r \geq 0.95$). We’re heading towards inference.
- Examine scatterplots to see the type of relationships. Non-linear patterns may suggest the correlation is strong at only some values (e.g., age 65+).
- Significant moderate and weak correlations are still useful. Even weak correlations can have a significant part to it (e.g., age 65+).

Course outline

Appendix A

Week 1

- **Distributions and correlations**
- **September 15, 2026/ 12:00-1:00 PST /MS Teams meeting**
- **Course Objectives:**
 - **Discuss and explain the background of descriptive statistics of distributions and correlations.**
 - **Demonstrate how to produce statistics and create graphs in statistical software.**

Week 2

- **Inference & Hypothesis Testing**
- **September 22, 2026/ 12:00-1:00 PST /MS Teams meeting**
- **Course Objectives:**
 - **Discuss and explain the background of inferential statistics and hypothesis testing.**
 - **Demonstrate how to produce statistics and create graphs in statistical software.**

Week 3

- **Power Analysis**
- **September 29, 2026/ 12:00-1:00 PST /MS Teams meeting**
- **Course Objectives:**
 - **Discuss and explain the background of power analysis for t-tests and chi-square tests.**
 - **Demonstrate how to produce statistics and create graphs in statistical software.**

Week 4

- **Survey Research Methods**
- **October 6, 2026/ 12:00-1:00 PST /MS Teams meeting**
- **Course Objectives:**
 - **Discuss and explain the background of Survey Research Methods.**
 - **Demonstrate how to produce statistics and create graphs in statistical software.**

Week 5

- **Linear & Logistic Regression**
- **October 13, 2026/ 12:00-1:00 PST /MS Teams meeting**
- **Course Objectives:**
 - **Discuss and explain the background of Simple and Multiple Linear and Logistic Regression.**
 - **Demonstrate how to produce statistics and create graphs in statistical software.**

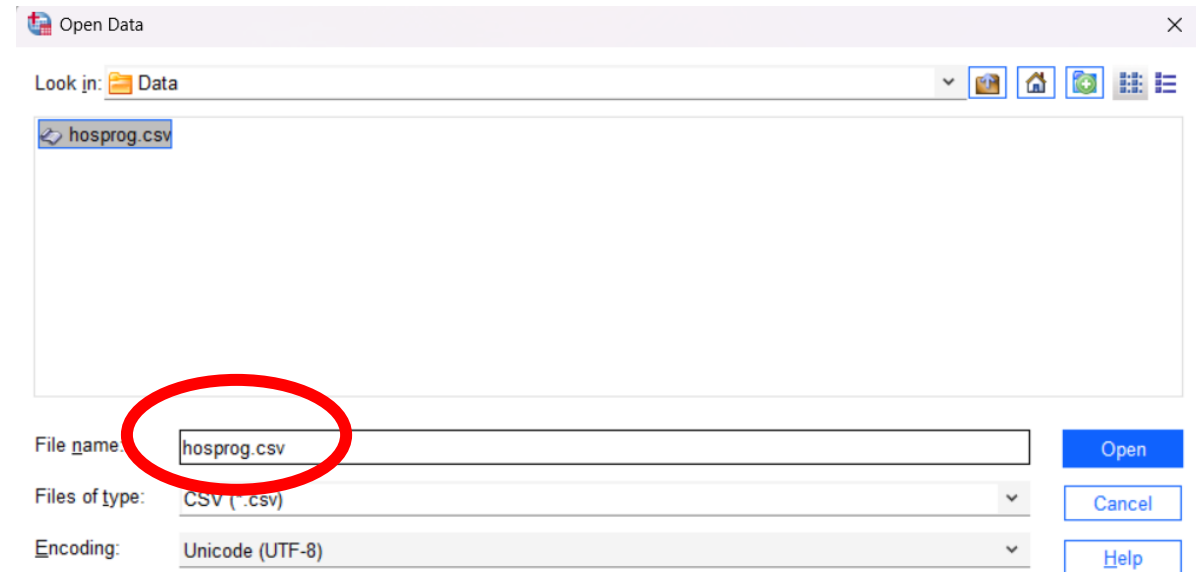
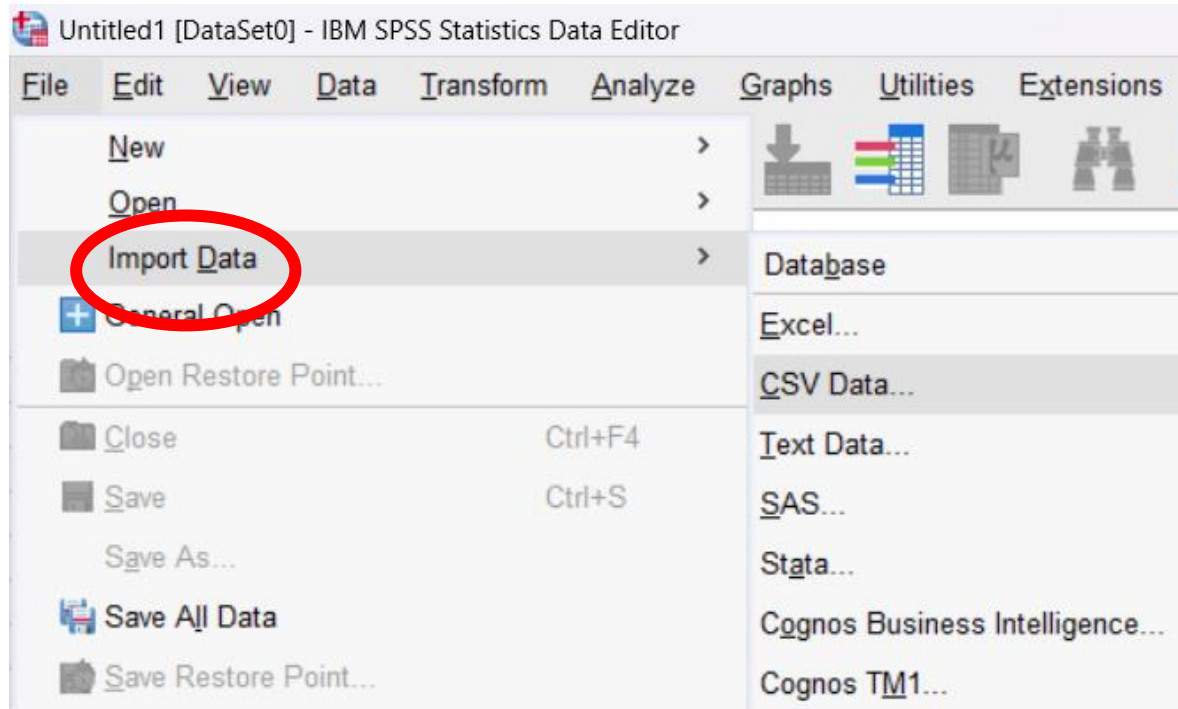
Week 6

- **Matching**
- **October 20, 2026/ 12:00-1:00 PST /MS Teams meeting**
- **Course Objectives:**
 - **Discuss and explain the background of propensity score matching for hypothesis testing and regression.**
 - **Demonstrate how to produce statistics and create graphs in statistical software.**

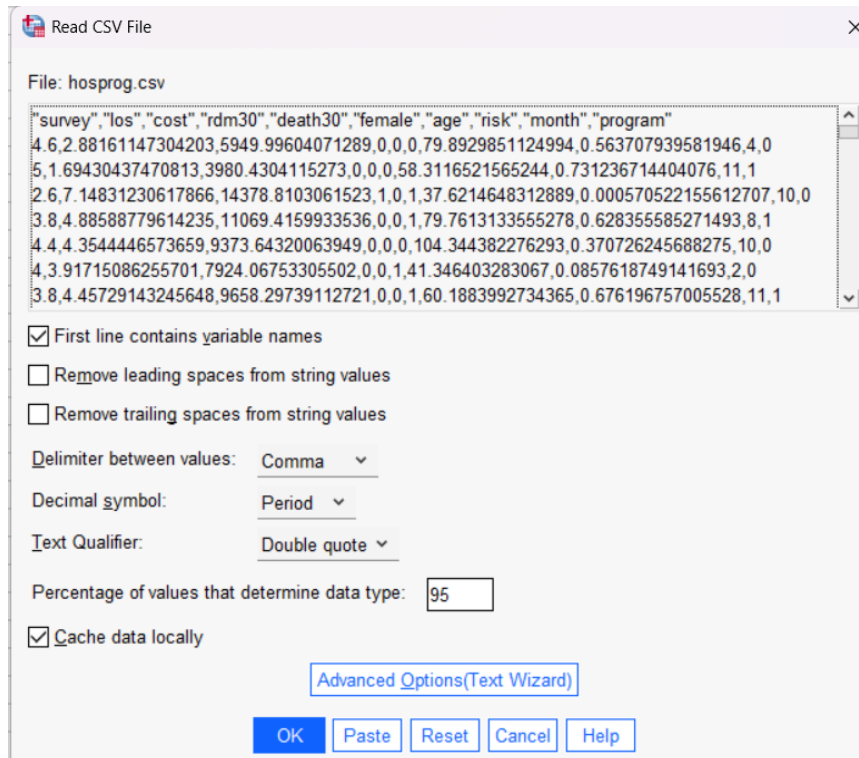
SPSS examples of how to import data, create a distribution graph, scatterplot, and correlation table

Appendix B

Import a CSV file



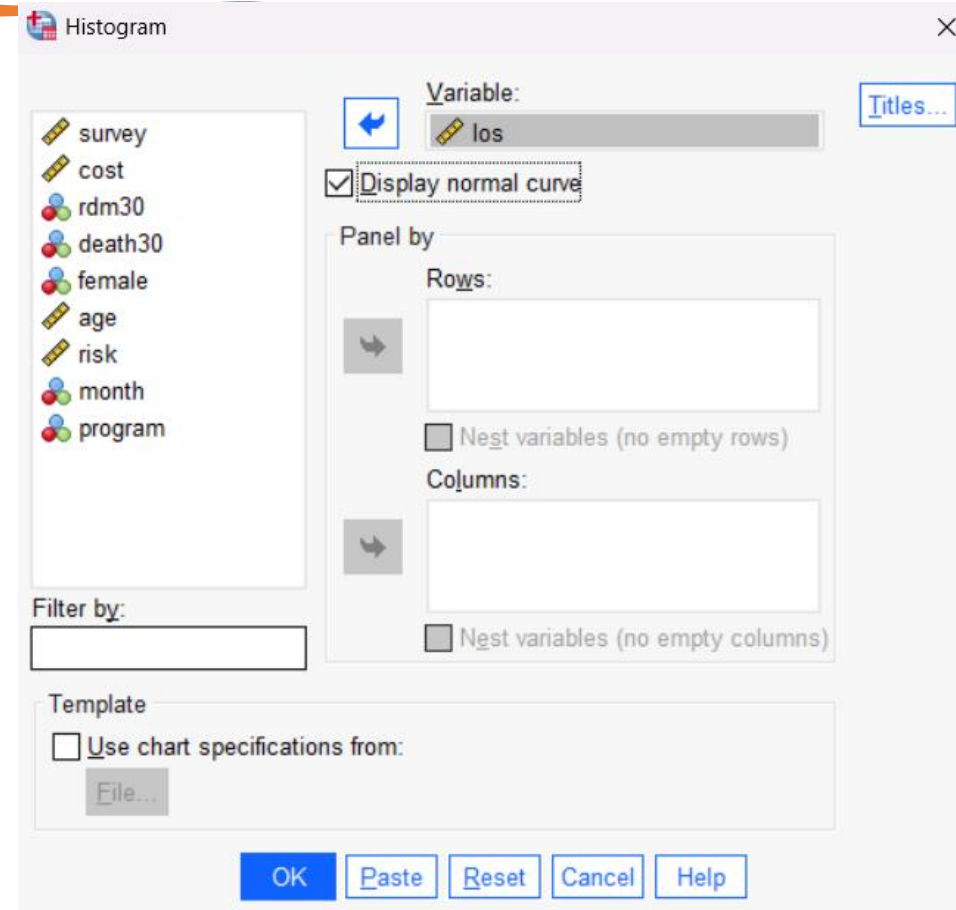
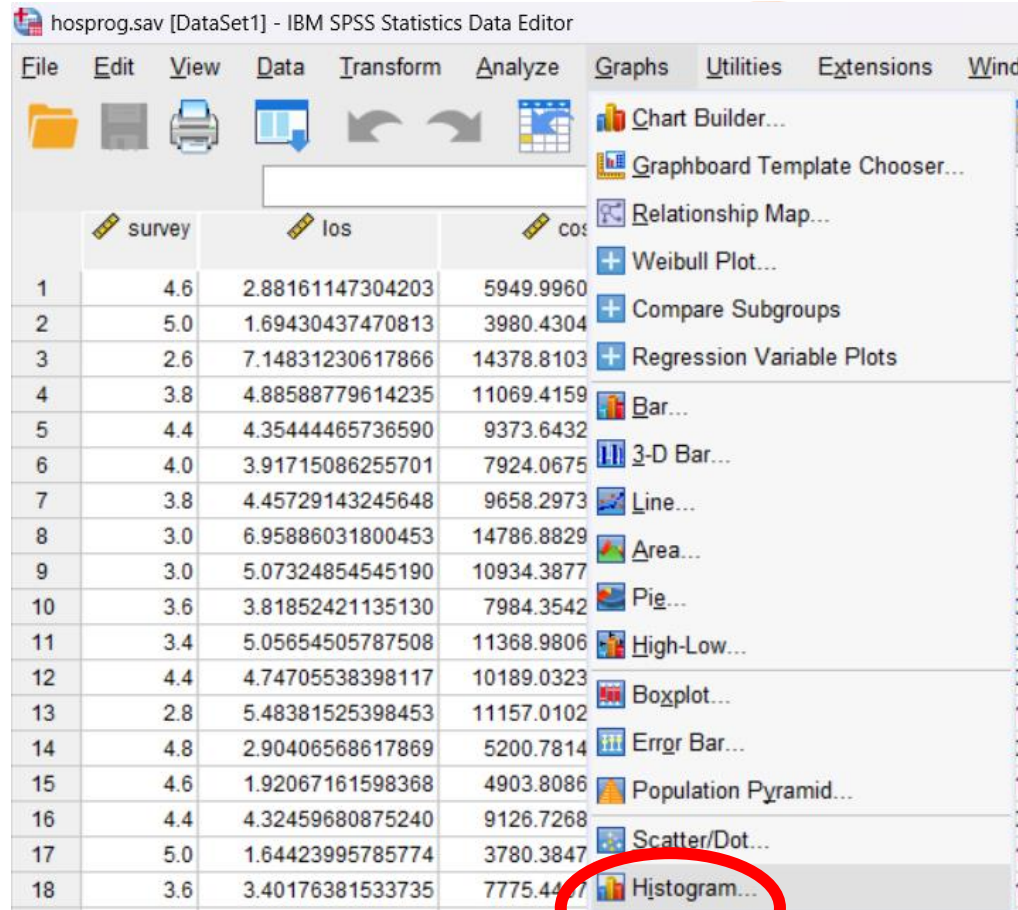
Format a CSV into a SPSS file



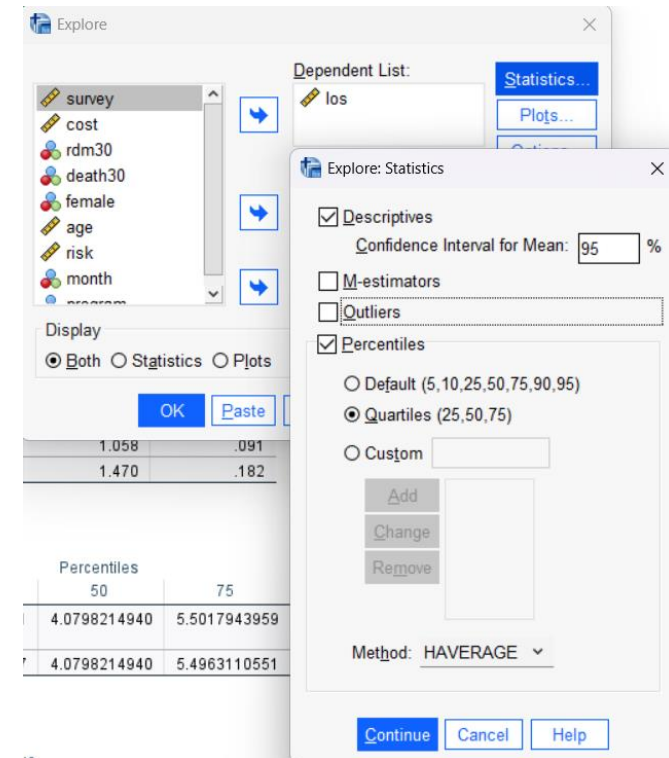
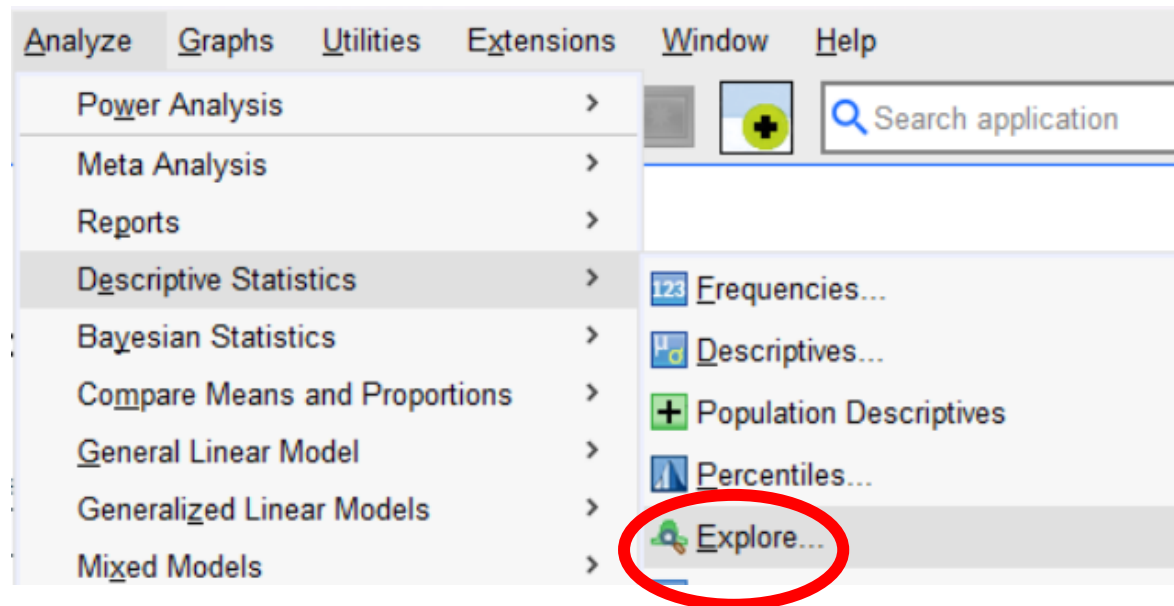
The screenshot shows the IBM SPSS Statistics Data Editor with the data from the CSV file loaded into a table. The table has 7 columns: survey, los, cost, rdm30, death30, female, and age. The data is organized into 6 rows, with the first row containing the variable names and the subsequent rows containing numerical values.

	survey	los	cost	rdm30	death30	female
1	4.6	2.88161147304203	5949.99604071289	0	0	0
2	5.0	1.69430437470813	3980.43041152730	0	0	0
3	2.6	7.14831230617866	14378.81030615230	1	0	1
4	3.8	4.88588779614235	11069.41599335360	0	0	1
5	4.4	4.35444465736590	9373.64320063949	0	0	0
6	4.0	3.91715086255701	7924.06753305502	0	0	1

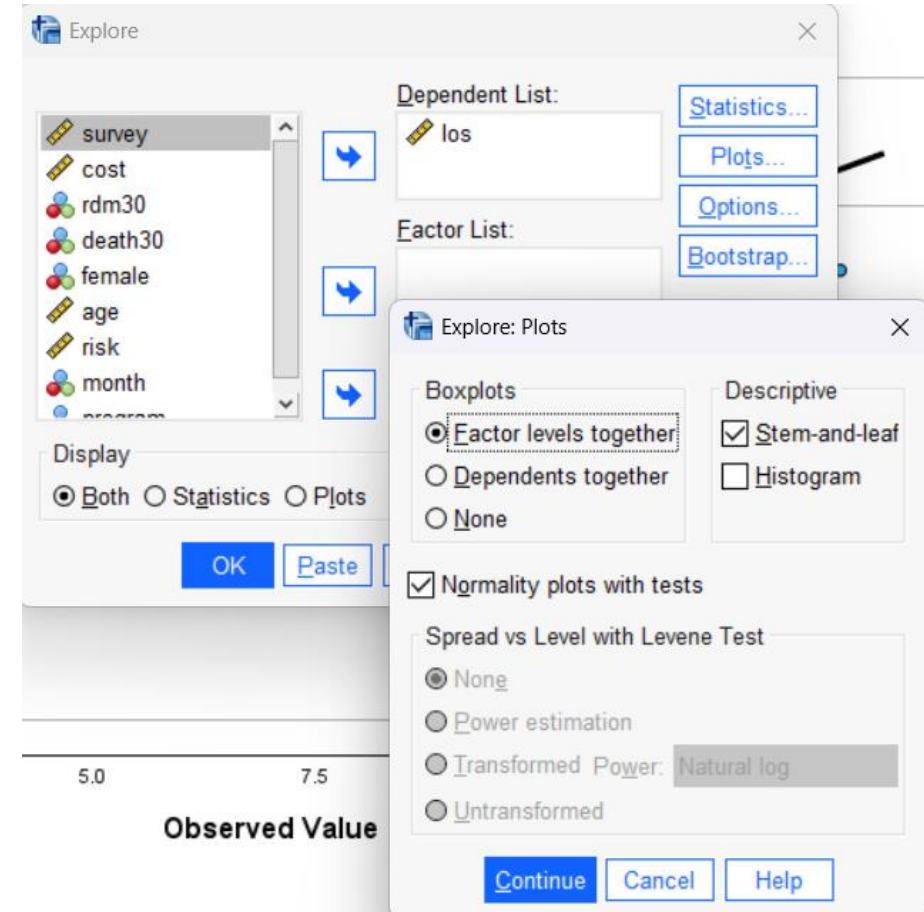
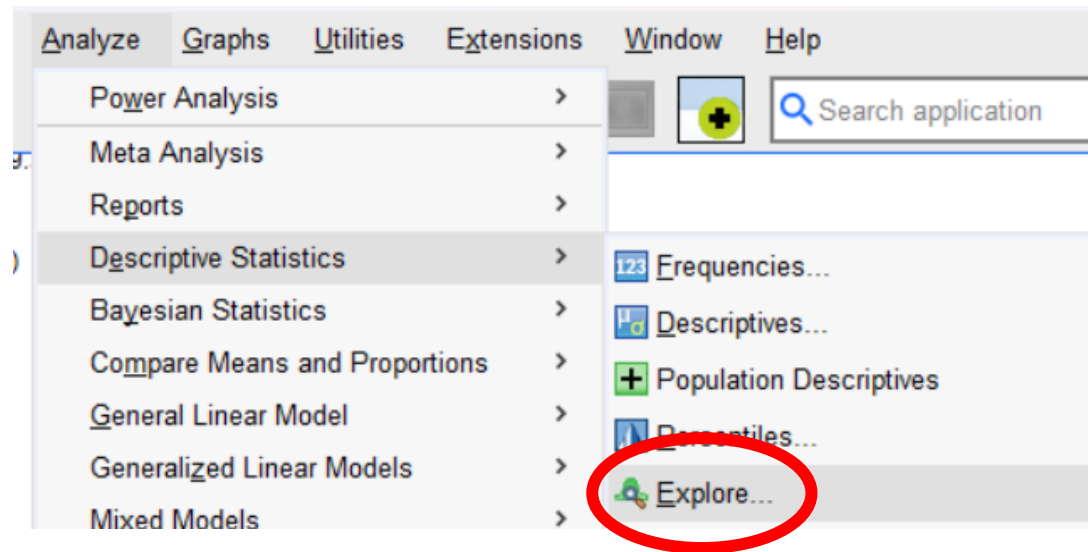
Creating a Histogram



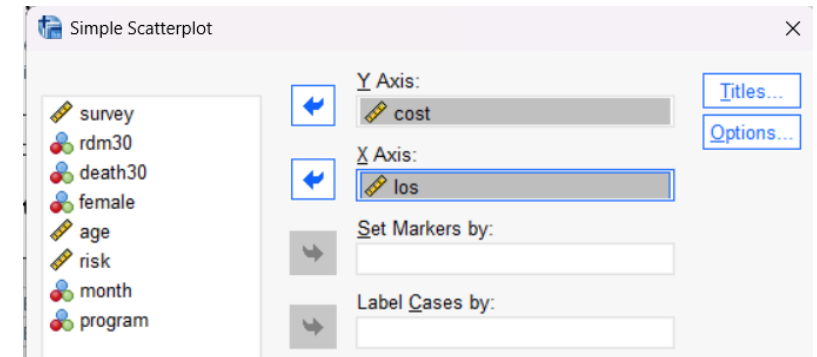
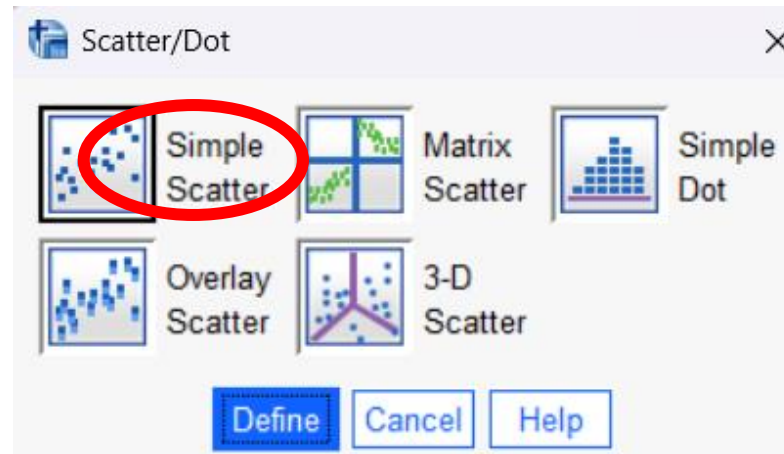
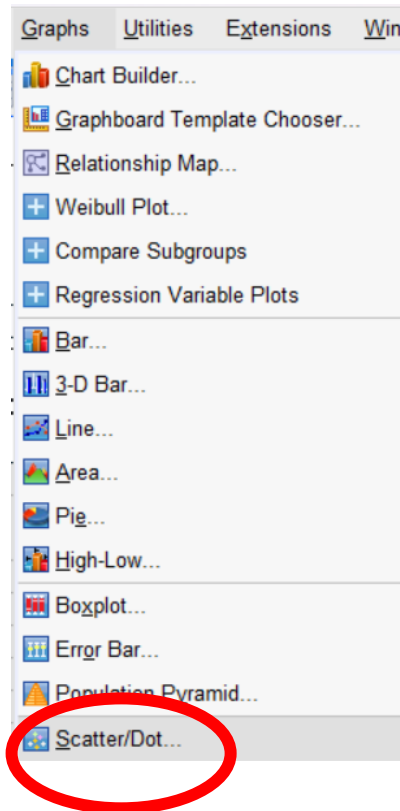
How to get Descriptive Statistics



How to create a Q-Q plot and run a Kolmogorov-Smirnov test

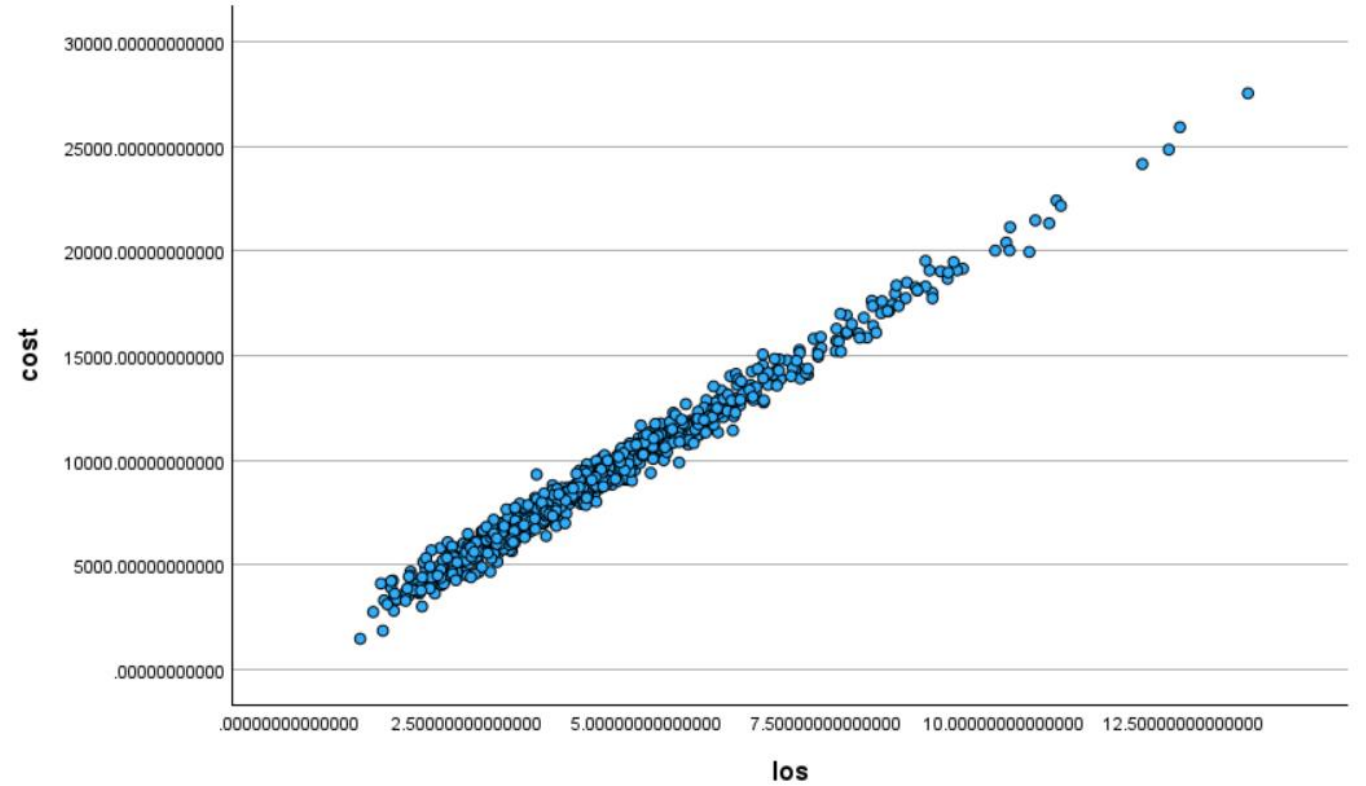


How to create a scatterplot

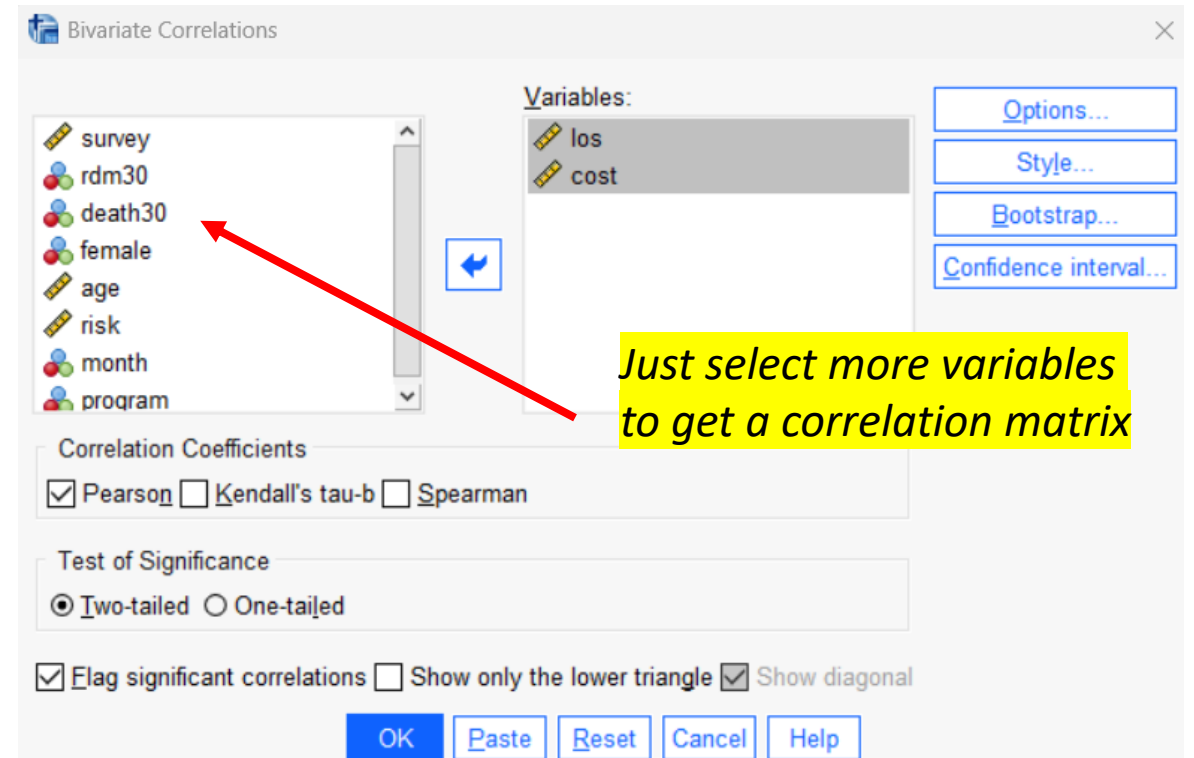
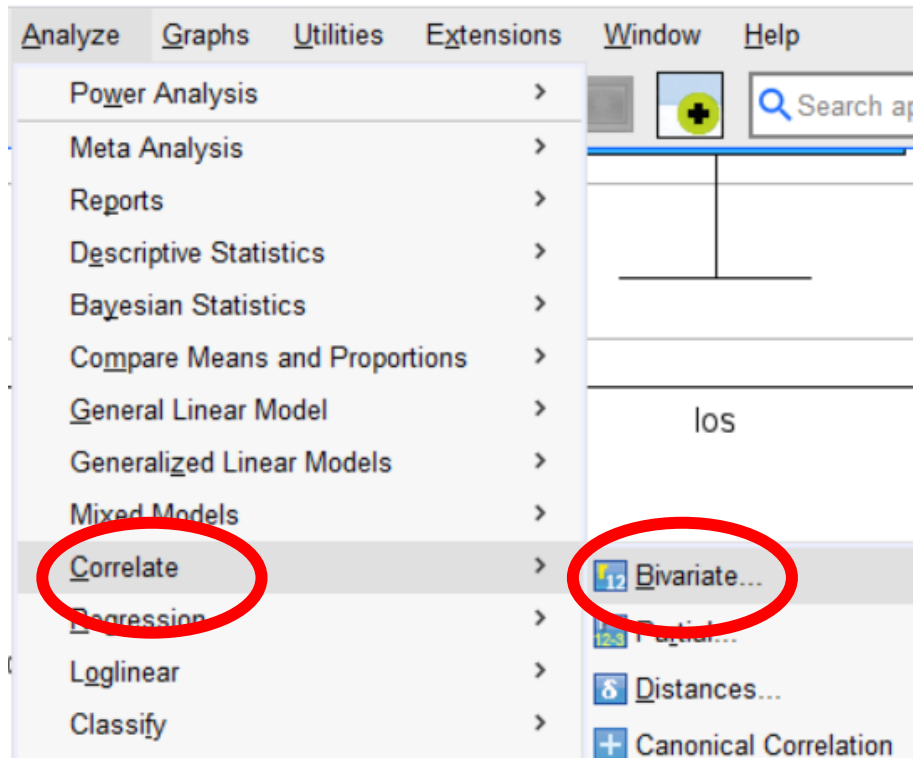


Scatterplot

Graph



How to run a correlation test



Correlation and statistical test

Correlations

		los	cost
los	Pearson Correlation	1	.992**
	Sig. (2-tailed)		<.001
	N	720	720
cost	Pearson Correlation	.992**	1
	Sig. (2-tailed)	<.001	
	N	720	720

** . Correlation is significant at the 0.01 level (2-tailed).